Velora: Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes Velora's customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide Velora's strategic business decisions.

2. Dataset Summary

- Rows: 3,900 - Columns: 18 - Key Features:
- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using **pandas**.
- **Initial Exploration:** Used **df.info()** to check structure and **.describe()** for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discou Appli
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	Na
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	Na
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	Na
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	Na
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	Na
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	Na
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	Na

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the **Review Rating** column using the median rating of each product category.

- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.

- **Feature Engineering:**

    - Created **age_group** column by binning customer ages.

    - Created **purchase_frequency_days** column from purchase data.

- **Data Consistency Check:** Verified if **discount_applied** and **promo_code_used** were redundant; dropped **promo_code_used.**

- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into Velora's database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer Velora's key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88

Total rows: 839 Query complete 00:00:0

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased text	average_product_rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	average_purchase_amount numeric
1	Standard	58.46
2	Express	60.48

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status text	total_customers bigint	average_spend numeric	total_revenue numeric
1	No	2847	59.87	170436.00
2	Yes	1053	59.49	62645.00

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	number_of_customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	2	Accessori...	Belt	161
4	3	Accessori...	Scarf	157
5	1	Clothing	Blouse	171
6	1	Clothing	Pants	171
7	2	Clothing	Shirt	169
8	3	Clothing	Dress	166
9	1	Footwear	Sandals	160
10	2	Footwear	Shoes	150
11	3	Footwear	Sneakers	145
12	1	Outerwear	Jacket	163
13	2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status text	repeat_buyers bigint	percentage numeric
1	No	2518	72.44
2	Yes	958	27.56

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group text	total_revenue numeric	revenue_percentage numeric
1	Young Adult	62143	26.66
2	Middle-aged	59197	25.40
3	Adult	55978	24.02
4	Senior	55763	23.92

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present Velora's insights visually.



6. Key Findings

- **Uniform Spending** – The average purchase is $59.76, and gender, age group, subscription status, discount use and shipping type all average within $2.30 of each other (items 1, 4, 5 and 10).

- **Revenue Follows Volume** – Men are 68% of customers and generate 67.7% of revenue (item 1). Clothing and Accessories bring 76.6% of revenue because they account for most purchases.

- **Subscription** – Only 27% of customers subscribe, generating 26.9% of revenue, and subscribers spend about the same per purchase as non-subscribers ($59.49 vs. $59.87, item 5). All 1,053 subscribers are male; none of the 1,248 female customers subscribe.

- **Mature Customer Base** – 79.9% of customers are Loyal (more than 10 previous purchases), 18.0% Returning and 2.1% New (item 7). Repeat buyers (89.1% of customers) subscribe at 27.6% vs. 22.4% for other customers (item 9).

- **Discounts** – 43% of purchases used a discount, yet discounted purchases average $59.28 vs. $60.13 at full price, and 839 of 1,677 discount users (50%) still spent above average (item 2). The five most discounted products (47–50%) sit only 4–7 points above the overall rate (item 6).

- **Small Gaps Elsewhere** – Top-rated products span only 3.78–3.86 (item 3), best-sellers in each category are within 15 orders of each other (item 8), Express averages $2.02 more than Standard (item 4), and Young Adults lead revenue (26.7%) mainly because of group size (item 10).

- **Data Notes** – Each record is one customer, so repeat behavior can only be read from previous purchases; the 37 missing ratings were estimated using category medians; and with no female subscribers, gender and subscription effects cannot be separated.

7. Business Recommendations

- **Boost Subscriptions** – Only 27% of customers subscribe, and none of the 1,248 female customers do (vs. 39.7% of men). Confirm whether this is an eligibility or data issue, then test an offer for women. Subscribers spend about the same per purchase, so promote benefits that drive repeat purchases and track repeat-purchase rate.

- **Customer Loyalty Programs** – 79.9% of customers already count as Loyal, so add reward tiers across the 1–50 range of previous purchases (quartiles: 13, 25, 38) and use higher tiers to nudge repeat buyers toward subscribing.

- **Review Discount Policy** – Discounts show no spending benefit ($59.28 vs. $60.13), and half of discount users spent above average anyway. Test reducing blanket discounts, starting with Hat, Sneakers, Coat, Sweater and Pants, and compare sales and margin.

- **Product Positioning** – Highlight best-sellers (Jewelry, Blouse, Pants, Sandals, Jacket) and top-rated products (Gloves, Sandals, Boots), focusing on Clothing and Accessories (76.6% of revenue). Test bundling Outerwear (7.9% of revenue, lowest average purchase at $57.17) with Clothing purchases.

- **Targeted Marketing** – Prioritize categories and customer volume over age or shipping segments: group averages differ by $2.30 or less, so Young Adults (26.7% of revenue) and Express users ($60.48 vs. $58.46 for Standard) are only modestly better targets.